

Number Theory: Addition

In this lab you've learned the basics of number theory as it relates to addition.

Rubric

Item	Description	Value
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Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Questions

Summary: In this lab we learned how to create and use basic adders as well as full adders. We also continued to practice coding in verilog and using k-maps to minimize equations. We got to see how logic gates combine to perform mathematical operations through binary.

1 - How might you add more than two bits together?

You can add more than two bits together by connecting multiple full adders together and making the carry out of one the carry in of the next. The more bits that are added require extra outputs since it's no longer just 0 or 1. If you continue the adders, the circuit can scale to handle binary numbers of any bit length.

2 - What is the importance of the XOR gate in an adder? The XOR gate in the adder detects when there is either a carry in 1 or a 1 input but not both. This allows us to calculate the sum of the full adder. The XOR gate will also output 1 for the sum if there are an odd number of inputs.

3 - What is the largest number a two bit adder can handle? What happens when you go over? The largest number a two bit adder can handle is 3, it can handle four values from 0 to 3. When you go over this all the values of the adder become 0, but the carry out becomes 1.